

Matthew Landers

mattlanders.net qwp4pk@virginia.edu (201) 417-6700

Research Focus	Developing structured and scalable reinforcement learning algorithms for high-dimensional decision-making, with applications to foundation models, agents, and real-world control systems.	
Education	University of Virginia , Charlottesville, VA	2021-2026
	Ph.D., computer science Advisors: Dr. Thomas Hartvigsen and Dr. Afsaneh Doryab GPA: 4.0 Dissertation: Reinforcement Learning in Combinatorial Action Spaces	
	Johns Hopkins University , Baltimore, MD	2019-2021
	M.S., computer science Advisor: Dr. Suchi Saria GPA: 4.0	
	Indiana University, Kelley School of Business , Bloomington, IN	2008-2012
	B.S., entrepreneurship and corporate innovation	
Publications	[1] Matthew Landers, Taylor Killian, Tom Hartvigsen, Afsaneh Doryab. “Improving and Accelerating Offline RL in Large Discrete Action Spaces with Structured Policy Initialization” ICLR, 2026. [2] Matthew Landers, Taylor Killian, Hugo Barnes, Tom Hartvigsen, Afsaneh Doryab. “BraVE: Offline Reinforcement Learning for Discrete Combinatorial Action Spaces” NeurIPS, 2025. [3] Matthew Landers, Taylor Killian, Tom Hartvigsen, Afsaneh Doryab. “SAINT: Attention-Based Modeling of Sub-Action Dependencies in Multi-Action Policies” <i>Under review</i>. [4] Matthew Landers, Afsaneh Doryab. “Parameter Transfer for Single-Task Reinforcement Learning.” International Joint Conference on Neural Networks, 2025. [5] Matthew Landers, Afsaneh Doryab. “Deep Reinforcement Learning Verification: A Survey.” ACM Computing Surveys, 2023. [6] Echo Wang*, Matthew Landers*, Roy Adams*, Adarsh Subbaswamy, Hadi Kharrazi, Darrell Gaskin, Suchi Saria. “A bias evaluation checklist for predictive models and its pilot application for 30-day hospital readmission models.” Journal of the American Medical Informatics Association, 2022. [7] Matthew Landers, Ray Dorsey, Suchi Saria. “Digital Endpoints: Definition, Benefits, and Current Barriers in Accelerating Development and Adoption.” Digital Biomarkers, 2021. [8] Matthew Landers, Suchi Saria, Alberto Espay. “Can Artificial Intelligence Replace the Movement Disorders Specialist for Diagnosing and Managing Parkinson’s Disease?” Journal of Parkinson’s Disease, 2021. [9] Chen Qian, Patraporn Leelaprachakul, Matthew Landers, Carissa Low, Anind K. Dey, Afsaneh Doryab. “Prediction of Hospital Readmission from Longitudinal Mobile Data Streams.” Sensors, 2021. * indicates equal contribution	
Technical Writing	Maintain long-form technical notes on reinforcement learning and optimization, emphasizing clarity, derivations, and practical algorithms on mattlanders.net	2021-PRESENT

Industry Experience	Head of Artificial Intelligence (Founding Team), Bay Compute	2025-PRESENT
	- Designed and deployed a safety-constrained offline reinforcement learning controller that reduced power consumption by 20% with zero operational constraint violations - Built rack-level testing schedule optimizer that increased throughput by 32–50%, improved average power utilization by 40–45% , and recovered 175 kW of peak capacity - Architected physics-augmented simulation environment of production data centers to model nonlinear thermal, airflow, and power dynamics from operational telemetry	
	Senior Software Engineer, Stride Consulting	2017-2019
	Client: Schonfeld Strategic Advisors - Built stateless Python services and stateful Java services to efficiently process billions of dollars in trades - Improved system latency from 1.1 seconds to 15 milliseconds and reduced system recovery time from 2 hours to less than 5 minutes by changing data transfer method from HTTP to Kafka	
	Client: Peloton Interactive - Led development of payment processing and order fulfillment tools for Australian launch - Designed and implemented an approach to facilitate communication with new Enterprise Resource Planner	
	Client: Magnetic - Rebuilt core platform growing monthly recurring revenue from \$1,000,000 to \$6,000,000 - Engineered Python microservices that allowed users to manage each component of an online advertisement including audience definition, budget selection, and media creation - Built microservices that published advertisements to 7 different networks	
	Software Engineer, Human API	2016-2017
	- Worked with data science and product teams to build integration solutions for 40+ customers - Engineered custom products for 30+ clients including a user health timeline for Samsung - Devised a partnership plan with executives from Merck Germany	
	Co-founder & CEO, Stonecrysus	2013-2016
	- Developed a machine learning health platform that earned 4 United States patents - Worked with an electrical engineering and design firm to manufacture a proprietary fitness wearable - Presented 3 times to groups of 300+ people, including at CES - Negotiated with organizations such as Samsung, Amazon, and the US Olympic Committee - Designed and built a web platform that allowed health practitioners to access and analyze patient data	
	Co-founder & President, Synduit	2012-2013
	- Grew monthly recurring revenue from \$0 to \$100,000+ in less than 1 year with a team of 2 people - Drove profit margin to 80% by managing construction of a machine learning-based automation platform - Managed 12 employee team while building client base to 120+ businesses	
	Intern, Tradeweb Markets	2010, 2011
Awards	Outstanding PhD Award, School of Engineering and Applied Sciences, University of Virginia	2026
	University of Virginia Endowed Graduate Fellowship	2024
Teaching	University of Virginia	
	Teaching Assistant, Artificial Intelligence	2023, 2024
	Teaching Assistant, AI for Social Good	2021
	Johns Hopkins University	
	Course Assistant, Data Structures	2019, 2020

Mentorship	University of Virginia Graduate School Mentor Program	2022
	UVA Computer Science Graduate Student Group Mentorship Program	2021
	Engineering Manager, Stride Consulting	2018-2019
	Stride Consulting Mentorship Program	2017-2019
Service	Reviewer	
	ICML	2026
	AISTATS	2025, 2026
	ICLR	2025, 2026
	NeurIPS	2024-2026
	ACM Computing Surveys	2024
	KDD	2024
	UbiComp	2022, 2023
	CHI	2023
	Student Volunteer	
	UbiComp	2021, 2022
Patents	us 9,317,815: Health and fitness management system	ISSUED 2016
	us 9,183,498: Health and fitness management system	ISSUED 2015
	us 8,892,481: Health and fitness management system	ISSUED 2014
	us 8,600,928: Health and fitness management system	ISSUED 2013